

Sai Vivek Padakanti

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EDUCATION

Colorado State University

Master of Science, Computer Science

Aug 2023 - Dec 2025

VNR Vignana Jyothi Institute of Engineering & Technology

Bachelor of Technology, Major: Electronics & Instrumentation Engineering, Minor: Artificial Intelligence & Machine Learning

Aug 2019 - May 2023

SKILLS

Programming Languages: Python, C, C++, JavaScript, SQL, HTML, CSS, Kivy, PLC Programming

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Matplotlib, Streamlit, OpenCV, TensorFlow, PyTorch, Seaborn, Bootstrap

Tools & Platforms: Jenkins, Git/GitHub, Jira, Tableau, Postman, Figma, Linux, MacOS, VSCode

Other: CI/CD, REST & WebSocket APIs, Data Visualization, Machine Learning

EXPERIENCE

Colorado State University

Jan 2025 - Present

Research Assistant *Python, Pandas, Scikit-learn, Google Apps Script, JavaScript, Google Forms API, Microsoft Graph API*

- Developed an NLP-powered recommendation engine that matched 85% of students to relevant university clubs based on survey responses.
- Aggregated data from over 500 survey responses, engineering student interest profiles from structured and open-ended inputs.
- Built a scalable recommendation pipeline using TF-IDF and cosine similarity, reducing manual matching effort by 90%.

Morgan Library, Colorado State University

Aug 2023 - Present

Peer Research Mentor *Research, Organizational, Communicational & Interpersonal Skills, Mentorship, Technical Proficiency*

- Mentored and trained 50+ students on academic research tools, databases, and citation management software.
- Conducted data analysis from library databases to identify trends in student engagement, improving workshop effectiveness.
- Boosted engagement with library databases and citation tools by 40% through tailored peer guidance.
- Delivered weekly workshops and 1:1 sessions, building interpersonal, technical writing, and presentation skills.

Crestron Electronics, Inc.

May 2024 - Aug 2024

Firmware Engineer *Python, Kivy, JSON, REST API, WebSocket API, CI/CD, Jenkins, Jira, SVN, GitHub, VSCode*

- Automated CI/CD pipelines with Jenkins, improving deployment efficiency by 30% and reducing build failures by 25%.
- Designed a Python-Kivy application for audio control, improving real-time data speed by 25% and user satisfaction by 30%.
- Conducted button characteristic studies across 5+ products, informing design changes that boosted quality by 20% and user satisfaction by 15%.

PUBLICATIONS

Bird Species Identification Using Speech Recognition

Nov 2023

Published in the 2023 2nd International Conference on Futuristic Technologies (INCOFT), (IEEE)

PROJECTS

Python Mutation Testing Tool Comparison

Oct 2024 - Dec 2024

Python, Pytest, Unittest, Mutpy, MutMut, Mutatest, VSCode, GitHub

- Developed and implemented a Python codebase to evaluate and compare mutation testing tools on functionality and performance.
- Automated test suite generation for robust code coverage, using mutation scores to assess effectiveness.
- Delivered insights on tool suitability for real-world Python projects, boosting testing efficiency and reliability.

Anomaly Detection For Android Applications

Jan 2023 - May 2023

Python, Machine Learning, Streamlit

- Built a machine learning model that identified anomalous behavior in Android apps with 90% accuracy using real usage data.

CERTIFICATIONS & ACHIEVEMENTS

- Won the hackathon conducted by USDA to predict USDA Beef Quality grades using Image Recognition. Apr 2024
- Earned Google UX Design Professional Certificate, covering user research, interaction, and visual design. Apr 2023
- Trained in SAR Data Processing at IIRS, ISRO, covering raster images, contour maps, and 3D visualizations. May 2023